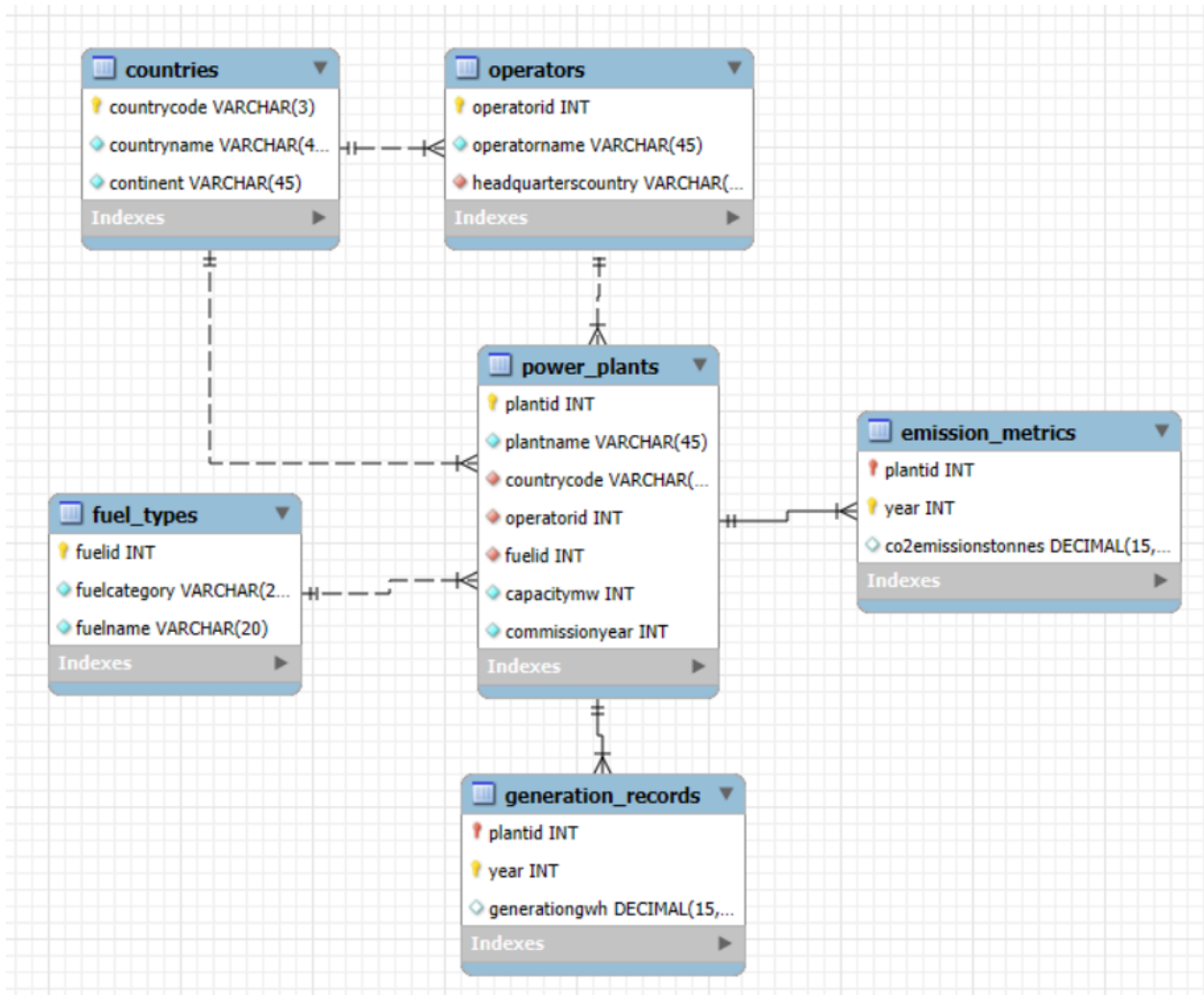


Title: Database Assignment 3

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Date: 10/6/2026



1. Retrieve the plant name, country name, operator name, fuel category, fuel name, capacity (in MW), and commission year for all power plants without using table aliases. Sort the results in descending order by capacity.

Select power_plants.plantname,
countries.countryname,
operators.operatorname,
fuel_types.fuelcategory,

	plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
▶	Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
	Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
	Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
	Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
	Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
	Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
	Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
	Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

```

fuel_types.fuelname,
power_plants.capacitymw,
power_plants.commissionyear
from power_plants
join countries on
power_plants.countrycode =
countries.countrycode
join operators on
power_plants.operatorid =
operators.operatorid
join fuel_types on
power_plants.fuelid =
fuel_types.fuelid
order by power_plants.capacitymw
desc;

```

- Retrieve the plant name, country code, calendar year, and annual generation (in GWh) for all power plants for the year 2024 without using table aliases. Sort the results in descending order by generation.

```

select plantname,
countrycode,
generation_records.year,
generation_records.generationgwh
from power_plants
join generation_records using
(plantid)
where generation_records.year =
2024
order by
generation_records.generationgwh
desc;

```

	plantname	countrycode	year	generationgwh
▶	Three Gorges	CHN	2024	98800.00
	Itaipu	BRA	2024	89000.10
	Tuocketuo	CHN	2024	71000.30
	Vindhyachal	IND	2024	35600.80
	Grand Coulee	USA	2024	21500.50
	Janschwalde	DEU	2024	18400.20
	Loy Yang	AUS	2024	14200.40
	Kashiwazaki-Kariwa	JPN	2024	0.00

3. Retrieve the plant name, country code, calendar year, annual generation (in GWh), and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases. Sort the results in ascending order by CO2 emissions.

```
select power_plants.PlantName,  
       power_plants.CountryCode,  
       generation_records.year,  
       generation_records.generationgwh,  
       emission_metrics.co2emissionstonnes  
from power_plants  
join generation_records  
    on power_plants.PlantID =  
       generation_records.plantid  
join emission_metrics  
    on power_plants.PlantID =  
       emission_metrics.plantid  
    and generation_records.year =  
       emission_metrics.year  
where generation_records.year =  
       2024  
order by  
       emission_metrics.co2emissionstonnes asc;
```

	PlantName	CountryCode	year	generationgwh	co2emissionstonnes
►	Grand Coulee	USA	2024	21500.50	0.00
	Itaipu	BRA	2024	89000.10	0.00
	Kashiwazaki-Kariwa	JPN	2024	0.00	0.00
	Three Gorges	CHN	2024	98800.00	0.00
	Loy Yang	AUS	2024	14200.40	1950000.00
	Janschwalde	DEU	2024	18400.20	2450000.00
	Vindhyachal	IND	2024	35600.80	3100000.00
	Tuoketuo	CHN	2024	71000.30	5200000.00

4. Write a SQL query using a **Common Table Expression (CTE)** to calculate the total cumulative power generation (in GWh) for each operator across all available years without using table aliases. Retrieve the operator name, headquarters country, and their total generated power, and sort the results in descending order by total generation.

```

with OperatorGeneration as (
    select
        power_plants.OperatorID,
        sum(generation_records.generationgwh
        ) as TotalGenerationGWh
    from power_plants
    inner join generation_records
        on power_plants.PlantID =
        generation_records.plantid
    group by
        power_plants.OperatorID
)
select
    operators.OperatorName,
    operators.HeadquartersCountry,

    OperatorGeneration.TotalGenerationGW
    h
from operators
inner join OperatorGeneration
    on operators.OperatorID =
    OperatorGeneration.OperatorID
order by
    OperatorGeneration.TotalGenerationGW
    h desc;

```

	OperatorName	HeadquartersCountry	TotalGenerationGWh
►	China Yangtze Power	CHN	169800.30
	Electrobras	BRA	89000.10
	NTPC Limited	IND	35600.80
	Global Hydro Inc	USA	21500.50
	Rhein Power AG	DEU	18400.20
	Snowy Hydro	AUS	14200.40
	Tepco Energy	JPN	0.00

- Write a SQL query using **two separate Common Table Expressions (CTEs)**--one to calculate total power generation by country code and another to calculate total CO2 emissions by country code without using table aliases. Then, join these CTEs with the countries table to display the country name, total generation (in GWh), and total CO2 emissions (in tonnes). Sort the results in descending order by total generation.

```

with power_country as
(
    select power_plants.CountryCode,
    sum(generation_records.generationgwh
) as gwhsum
    from power_plants
    inner join generation_records
        on power_plants.plantid =
generation_records.plantid
    group by
power_plants.CountryCode
),
country_emissions as
(
    select power_plants.CountryCode,
    sum(emission_metrics.co2emissionston
nes) as emsum
    from power_plants
    inner join emission_metrics
        on power_plants.plantid =
emission_metrics.plantid
    group by
power_plants.CountryCode
)
select countries.CountryName,
country_emissions.emsum,
power_country.gwhsum
from countries
inner join power_country
    on countries.CountryCode =
power_country.CountryCode
inner join country_emissions
    on countries.CountryCode =
country_emissions.CountryCode
order by power_country.gwhsum desc;

```

	CountryName	emsum	gwhsum
▶	China	5200000.00	169800.30
	Brazil	0.00	89000.10
	India	3100000.00	35600.80
	United States	0.00	21500.50
	Germany	2450000.00	18400.20
	Australia	1950000.00	14200.40
	Japan	0.00	0.00

